Big Data-Driven Commuter Advisory System

Big Data for Engineering – EEET2574

Team Members

Khoi Phan Anh

S3980639
Team Leader - Data Scientist

Hoang Tran Minh

S3975086
Visualization Specialist

Hai Nguyen Pham Tien

S3979239
Data Engineer

Chi Bui Hoang Quynh

S3978316
Data Engineer

Yen Nguyen Thi Hoang

S3930002
Data Scientist

Table of contents

01 Introduction to Project

02 Problem & Solution

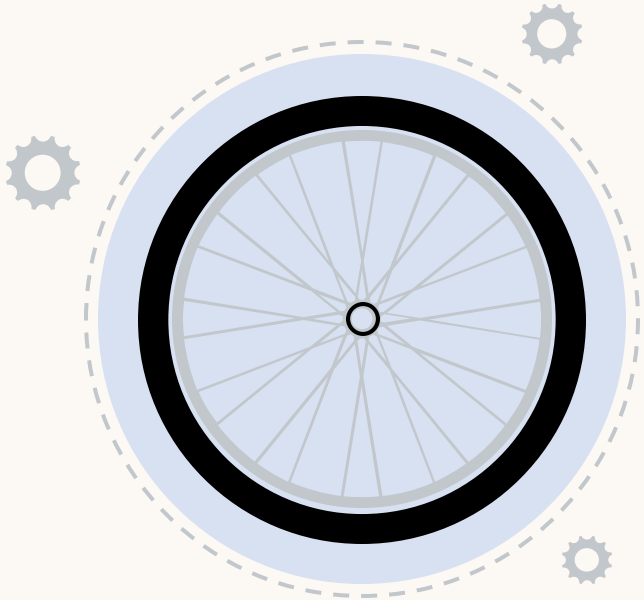
03 Datasets & Data Pipeline

04 Cost Estimation

05 Conclusion & Future Work

01

Introduction



What is our idea?

A **real-time commuter advisory system** that tells Londoners whether biking is a healthy, safe option.

Customer Value:

- **Simple & Personalized** – clear guidance, not raw data
- **Healthier Commuting** – reduce pollution exposure, bike safely
- **Time-Saving** – avoid disruptions and plan better routes

02

Problem & Solution

Project Identification

26.1M+

Trips on an average day in London

x2

Cycling trips in the last decade

-> Growing demand for sustainable mobility solutions

[1] “Travel in London 2024 Consolidated estimates of total travel and mode shares,” 2024. Available: https://content.tfl.gov.uk/travel-in-london-2024-consolidated-estimates-of-total-travel-and-mode-shares-acc.pdf

What Problem Are We Solving?

Problems

- Uncertainty about whether cycling is safe or healthy
- Weather, pollution, and road issues complicate decisions in London
- Current apps show raw data, not tailored advice

Solutions

- ML gives graded recommendations: **Safe / Moderate / Unsafe** for biking
- Visual dashboards with live stats
- AI provides simple, natural-language guidance

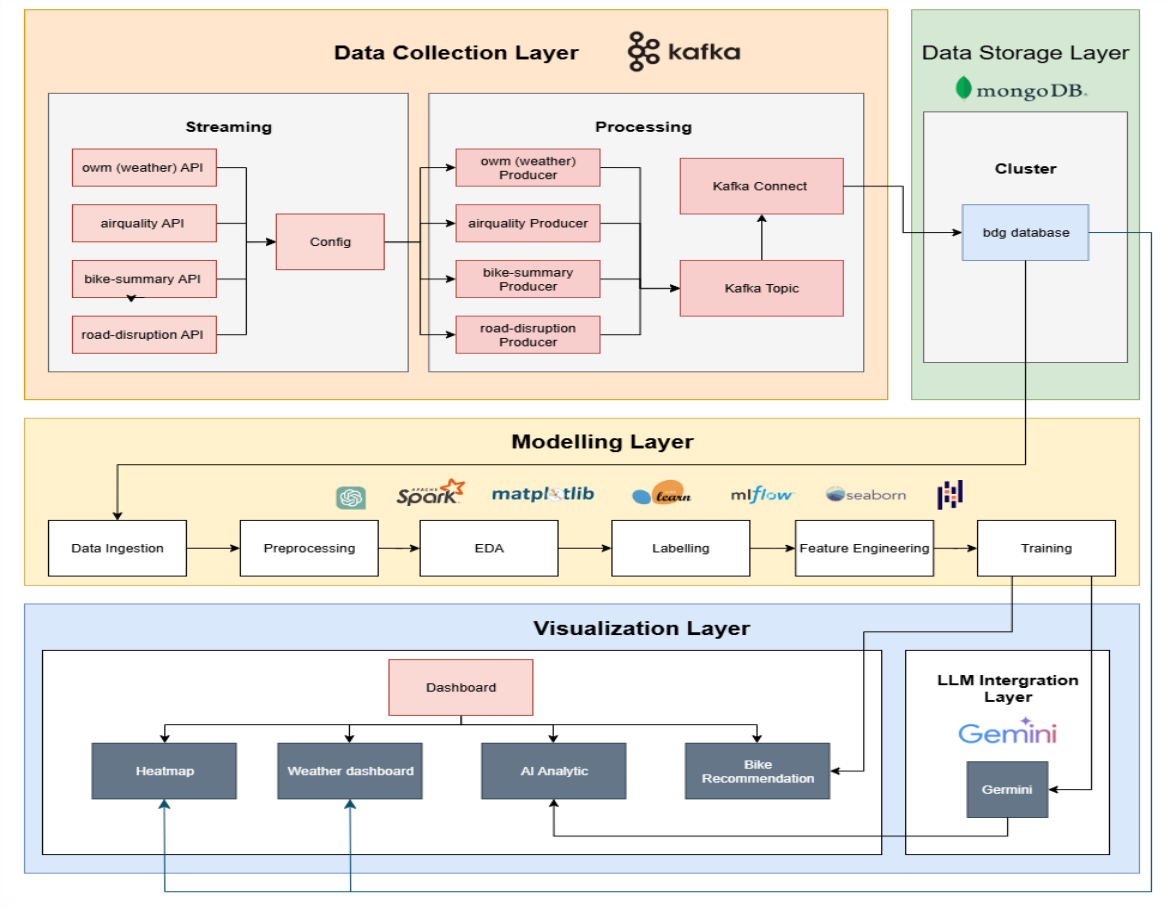
03

Datasets & Data Pipeline

Datasets

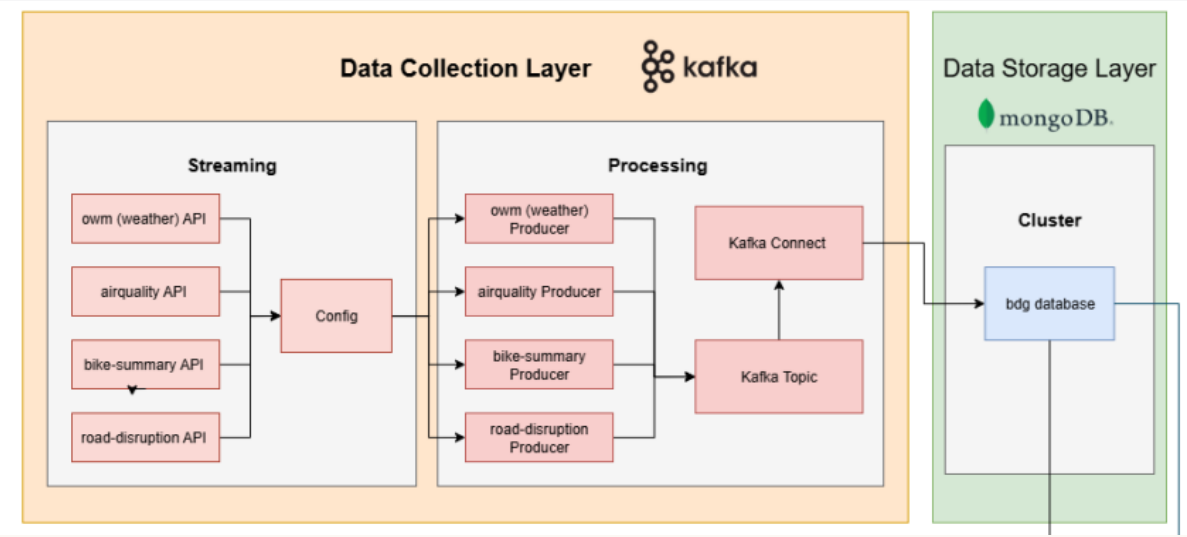
Dataset	Description	Key Fields	Usage
City Bikes API	Real-time bike availability at docking stations	station ID, location, available bikes, empty slots	Check nearby bike availability
OWM Weather API	Current weather conditions	temperature, humidity, rainfall, wind speed	Detect unsafe cycling conditions
OWM Air Quality API	Real-time air quality data	PM2.5, NO2, O3, and AQI index	Protect commuters from unhealthy air
TfL Road API	Current or scheduled road disruptions	road name, lat, lon , severity, active status	Avoid closed or disrupted streets

Data Pipeline



Data Collection & Storage

- Collected data from public APIs
- Ingests data from 4 producers into Kafka
- Stream into MongoDB across 4 collections:
  - `air_quality`
  - `bike_summary`
  - `weather`
  - `road_disruption`



Data Modelling Layer

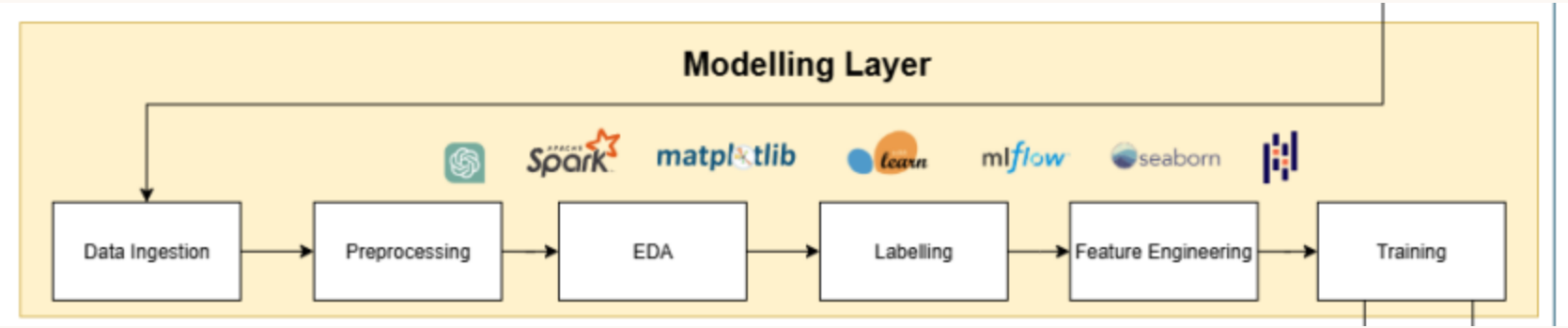
Preprocessing:

- Timezone harmonization (unified timestamp for London)
- 15-min resampling & synchronization
- Add temporal features (weekday/weekend)

EDA:

- Observed skewness, target imbalance, correlation.
- Applied Feature Triage Strategy to tag each feature as USE / EXPLORE / DELETE

Labeling: utilizing gpt5-nano model



EDA Details

Class Distribution – Target Label:

- Target variable has 3 classes: Good, Moderate, Unsafe
- Mild imbalance -> Consider class balancing methods like: SMOTENC or class weighting

Feature Triage Strategy:

- Applied 3 checks on numeric features: Constant / near-constant test, Skewness & kurtosis analysis, Mutual information with label

	Action	MI	skew	std
visibility	DELETE	0.000000	0.000000	0.000000
snow_1h	DELETE	0.060635	0.000000	0.000000
lon	DELETE	0.006156	0.000000	0.000000
aqi	DELETE	0.000000	0.000000	0.000000
lat	DELETE	0.051935	0.000000	0.000000

Feature Engineering

Filter Method

Variance Threshold + Mutual Information.

Embedded Method

Extra Trees importances.

Voting rule (≥2 methods)

→ Final feature set

Training

Split/Imbalance

Stratified train/test; in-fold oversampling hold + Mutual Information.

Models

RandomForest, ExtraTrees, GradientBoosting, HistGradientBoosting.

Cross Validation

5-fold GridSearchCV, scoring F1-macro.

ML Flow

Track and store modelling history.

Cross-validation Results

Model	F1-macro	Training time
Hist Gradient Boosting	0.939	2.21s
ExtraTrees	0.935	3.46s
Random Forest	0.925	8.37s
Gradient Boosting	0.916	2.41s

-> Final Model: Hist Gradient Boosting

Best Model Retrain:

Metrics	Value
Accuracy	0.8723
Balanced_accuracy	0.8611
F1_macro	0.8651
F1_weighted	0.8703
Precision_marco	0.8865
Recall_marco	0.8611
Roc_au_macro_ovr	0.9668

Imbalance handling:

Balanced Acc ≈ 0.86, Macro F1 ≈ 0.87 → Not majority-class biased.

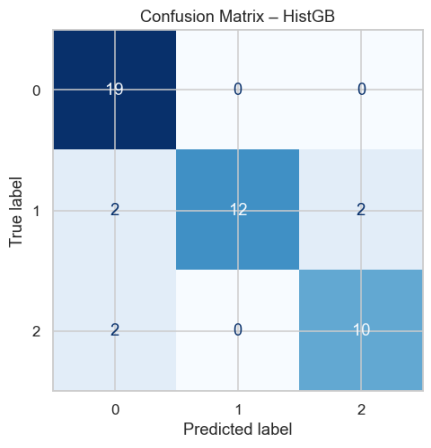
Class separability:

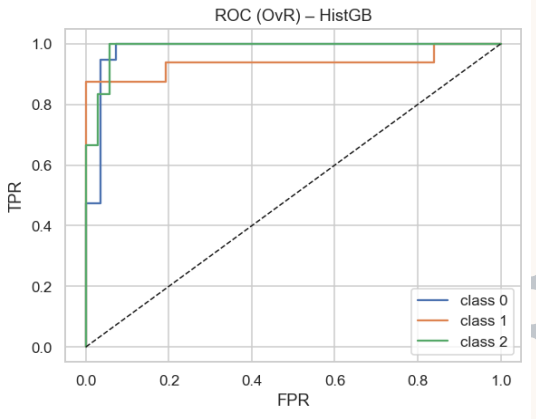
Macro ROC–AUC 0.967 → Clear separation and reliable decision boundaries.

Model quality

Overall: Accuracy & Macro F1 ≈ 0.87 → balanced, not majority-class biased.

Confusion Matrix: Good & Unsafe: strongly recognized; Moderate: main source of confusion.





Classification Report:

- ***Good & Unsafe***: Correctly flagged for most of the data points.
- ***Moderate***: Precision 1.00, Recall 0.75 → correct when predicted; some Moderates missed.

ROC–AUC: Good/Unsafe curves high; Moderate slightly lower → most boundary overlapping.

Classification Report:

	precision	recall	f1-score	support
0	0.83	1.00	0.90	19
1	1.00	0.75	0.86	16
2	0.83	0.83	0.83	12
accuracy			0.87	47
macro avg	0.89	0.86	0.87	47
weighted avg	0.89	0.87	0.87	47

ML Flow

bike-safety ›

Final-HistGB

Overview | Model metrics | System metrics | Traces | Artifacts

Description

No description

Details

Created at	09/11/2025, 02:51:51 PM
Created by	ADMIN
Experiment ID	2
Status	Finished
Run ID	40c44a1bea434893a928d37b8f8f5be3
Duration	3.5s
Datasets used	—
Tags	Add tags
Source	c:\Users\ADMIN\.conda\envs\bigdata311\Lib\site-packages\ipykernel_launcher.py
Logged models	—
Registered models	—
Registered prompts	—

Metrics (10)

Search metrics

Metric	Value
cv_f1_macro	0.9388531669788623
cv_time_sec	2.21
test_accuracy	0.8723404255319149
test_balanced_accuracy	0.8611111111111112
test_f1_macro	0.8650793650793651
test_f1_weighted	0.8703140830800405
test_precision_macro	0.8864734299516909
test_recall_macro	0.8611111111111112
test_cohen_kappa	0.8036211699164345
test_roc_auc_macro_ovr	0.9668404883175681

Parameters (21)

Search parameters

Parameter	Value
early_stopping	auto
interaction_cst	None
l2_regularization	0.0
learning_rate	0.1
loss	log_loss
max_bins	255
max_depth	None
max_features	1.0
max_iter	100
max_leaf_nodes	31
min_samples_leaf	20
monotonic_cst	None
n_iter_no_change	10
random_state	42
scoring	loss
tol	1e-07
validation_fraction	0.1
verbose	0
warm_start	False

Model Summary:

Strength:

- Excellent separability across all classes (macro AUC ≈ 0.97)
- Strong ability to flag Unsafe and Safe cases (critical for safety)
- Resilience to class imbalance

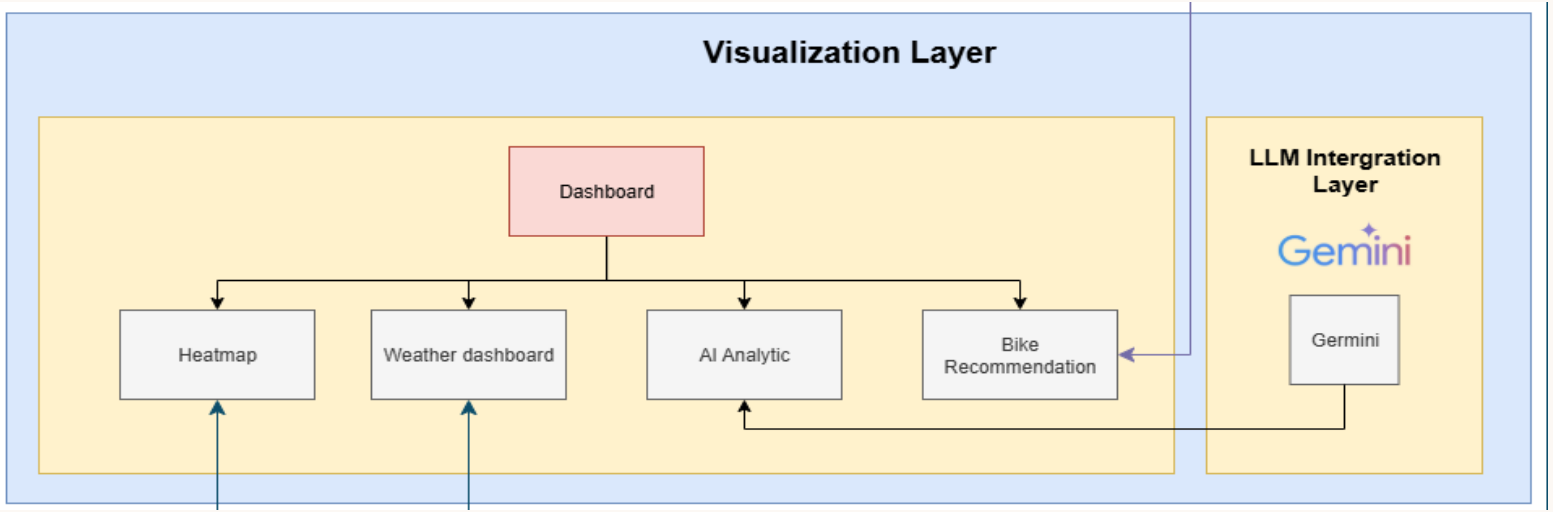
Weaknesses:

- Moderate class confuse with Good/Unsafe classes.

Data Visualization Layer

Dashboard Pages

- **Main**: Live weather, air quality & bike charts + auto/refresh options
- **Analytics**: AI-driven cycling insights
- **Recommendation**: ML-based bike advice classification
- **Heat Map**: Spatial disruption & condition view



04

Cost Estimation

Cost of Scaling Up

Service	Monthly Cost	Annual Cost	% of Total
MongoDB Atlas	$67	$804	1.1%
Gemini 2.5 Flash	$6,300	$75,600	98.9%
GPT 5 Nano	$0.056	$0.672	0.00088%
TOTAL	**$6,367**	**$76,404**	**100%**

$6,367/month for 20,000 users with 5× daily AI usage is extremely reasonable for a premium, AI-powered service

05

Conclusion & Future Work

Limitations

Achievements

- Integrated multiple live data sources
- Designed a scalable big data pipeline
- Delivered a working prototype with dashboard, classifier, and natural-language summaries

Limitations

- Current model: a simple classifier
- Limited training data available

Future Work

- Extend to multimodal transport
- Collect more data
- Expand system to other cities
- Improve prediction accuracy
- Add mobile push notifications

Thanks!

Do you have any questions?